

Identifying Sarcastic Spans

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Abstract

Sarcasm detection remains an important unsolved challenge in natural language processing. Failure to detect sarcasm can cause negative statements to be misclassified as positive sentiment, thereby distorting the identification of the intended tone. To investigate the most effective strategy for sarcasm detection, we conducted a comprehensive study comparing explicit manual feature engineering, such as dual-channel architectures, with implicit contextual representation learning. Overall, we evaluated performance across five distinct models: a baseline model (TF-IDF + SVM), a custom Dual-Channel Multi-Scale Convolutional Neural Network (DMS-CNN) that incorporates a semantic channel and a rule-based sentiment (VADER) channel, a hybrid BERT-CNN, and two Transformer models (BERT and DeBERTa).

Our results show that explicit architectural attempts fail to capture sarcasm. The custom DMS-CNN achieved only 69% accuracy, which is even lower than the simpler TF-IDF baseline (74%). In contrast, removing the manually designed structure and relying solely on end-to-end fine-tuning of DeBERTa produced the highest performance, reaching 82% accuracy. These findings suggest that the deep contextual attention mechanisms of pre-trained language models are substantially more effective for sarcasm detection than complex manually designed architectural channels.

1 Introduction

Sarcasm is a sophisticated linguistic phenomenon where the intended meaning of an utterance deviates from its literal interpretation. Linguistically, sarcasm is often grounded in context incongruity—a clash between positive lexical choices and negative situational contexts [3]. While recent advancements in Deep Learning have improved text classification, standard architectures often struggle to explicitly capture these incongruities, especially when the conflicting elements are separated by long-range dependencies. In this project, we focus on detecting sarcasm in informal text-only environments, such as social media posts and online forum discussions, where traditional conversational cues such as tone of voice and facial expressions are entirely absent.

Addressing this problem is critical for modern text analytics. Sentiment analysis is now widely used by companies for brand monitoring, by political an-

alysts to measure public opinion, and by automated customer service systems to process user feedback. However, sarcasm frequently disrupts these systems because sarcastic expressions can cause negative statements to be misclassified as positive sentiment. Accurate sarcasm detection therefore enables organizations to extract more reliable sentiment information and improve downstream decision-making.

Existing approaches generally fall into two categories: (1) feature-engineering methods, which rely on manually designed linguistic patterns, and (2) black-box neural models, such as standard LSTMs or Transformers, which learn dense representations but often lack explicit inductive biases for detecting structural irony. In addition, although multimodal models such as AMuSeD [2] leverage audio signals to capture sarcastic intonation, they are not applicable in text-only scenarios such as social media comments or forum discussions.

Our approach differs in that we do not propose only a single model; instead, we conduct a systematic evolutionary ablation study across three modeling paradigms. We first designed a novel architecture—the Dual-Channel Multi-Scale Convolutional Neural Network (DMS-CNN). By combining dual input channels (semantic and sentiment) with multi-scale convolutional filters, the model is intended to explicitly capture both local and long-distance discrepancies that characterize sarcasm. We then systematically compare this explicit, hand-crafted architecture with both traditional baselines and the implicit contextual learning of large pre-trained language models.

To evaluate these paradigms, we track performance progression across four distinct models: a traditional machine learning baseline (TF-IDF + SVM), our custom DMS-CNN, a hybrid BERT-CNN, and two fully fine-tuned Transformer models (BERT and DeBERTa). Our results show that the custom-designed DMS-CNN, achieving approximately 69% accuracy, underperforms the simpler TF-IDF baseline, which reaches 74%. However, by progressively removing manual feature engineering and ultimately fine-tuning DeBERTa end-to-end, performance improves to a peak accuracy of 82%.

The main contribution of this project is the insight that deep contextual modeling is substantially more important than manual feature engineering for sarcasm detection. Static, rule-based sentiment lexicons such as VADER can actively introduce noise, because they assign sentiment scores without considering surrounding context and therefore oversimplify the emotional reversals inherent in irony. As a result, resolving the complex contextual incongruities that define sarcasm is more effectively achieved by leveraging the deep, dynamic attention mechanisms of pre-trained language models rather than explicitly forcing a model to detect sentiment clashes through complex architectural channels.

2 Data

For this study, we utilize data from the University of California, Santa Cruz (UCSC) Sarcasm Corpus V2 [5]. The corpus is divided into three subsets. The largest is the generic sarcasm subset (GEN), which contains 3,260 posts

per class (sarcastic and non-sarcastic), totaling 6,520 posts. The hyperbole (HYP) and rhetorical question (RQ) subsets are smaller, containing 582 and 851 posts per class, respectively. Notably, some posts in the HYP and RQ subsets overlap with the GEN corpus. Specifically, HYP posts feature cue words that signal hyperbole, while RQ posts consist of question-answer structures where the speaker continues their own turn rather than allowing a direct response.

To maximize the volume of training data, we combined all three subsets into a single comprehensive dataset for model training and analysis. The dataset is structured with two primary columns: text, which contains the detailed content of the post, and class, a binary label indicating whether the post is sarcastic (sarc) or non-sarcastic (non-sarc). For example, a non-sarc instance from the GEN subset reads: "If that's true, then Freedom of Speech is doomed. Harassment is subjective. Now I can claim that a book I don't like is harassing me, and it will be banned."

3 Related Work

1. **Azwar, Azika & Suharjito, Suharjito. (2020), *Sarcasm Detection Using Multi-Channel Attention Based BLSTM On News Headline*. [1]:** The model adopts a Multi-Channel Attention-Based Bidirectional Long Short-Term Memory (BLSTM) network, and was evaluated on the News Headlines Dataset for Sarcasm. This paper enabling us to examine how information from two channels is integrated and contributes to the final prediction. In the original paper, the model utilizes GloVe and fastText embeddings, whereas our work will focus on CNN-based representations, allowing us to compare our results directly with those reported in the paper.
2. **Joshi et al. (2015), *Harnessing Context Incongruity for Sarcasm Detection* [3]:** Their main aim was to use SVM to prove that sarcasm is fundamentally a clash (incongruity) between two types of sentiment. We utilize the theoretical framework of context incongruity defined in this paper to guide our architectural design, specifically the choice of filter sizes to capture conflicting polarities.
3. **Kim (2014), *Convolutional Neural Networks for Sentence Classification*[4]:** This seminal work demonstrated that CNNs can effectively capture local features (n-grams) in text. We adopt this backbone but extend it significantly by introducing dual channels and attention mechanisms.
4. **Poria et al. (2016), *A Deeper Look into Sarcastic Tweets Using Deep Convolutional Neural Networks*[6]:** This work established the viability of CNNs for sarcasm detection. It used pre-trained CNNs to extract sentiment, emotion, and personality features from tweets and prove

that sarcasm is often a combination of specific emotional states (like frustration or mock joy) and sentiment flips. However, their approach relies on a single input channel. We improve upon this by explicitly modeling the interaction between semantic meaning and sentiment polarity.

5. **Liming Zhou, Xiaowei Xu, Xiaodong Wang (2023), *BNS-Net: A Dual-channel Sarcasm Detection Method Considering Behavior-level and Sentence-level Conflicts* [9]:** The paper proposes a dual-channel sarcasm detection model called BNS-Net, consisting of a Behavior-level Conflict Channel and a Sentence-level Conflict Channel to overcome the limitations of single-stream models. Since the Sentence-level Conflict Channel operates similarly to our Sentiment Channel B, this provides an opportunity to compare their dual-channel design with ours and evaluate how differences in Channel A influence overall prediction performance.

By synthesizing the methodologies from these foundational works, we have refined our dual-channel approach to explicitly address context incongruity. We utilize the multi-scale CNN framework established by Kim and Poria et al. as a structural backbone, while extending it with a sentiment-specific channel to capture the ‘polarity flips’ central to sarcastic expression. Furthermore, these papers provide critical performance benchmarks (Accuracy and F1-score) against which we can evaluate the DMS-CNN, allowing us to rigorously measure our model’s ability to identify sarcastic spans relative to existing state-of-the-art architectures.

4 Methodology

4.1 DMS-CNN

Dual-Channel Multi-Scale Convolutional Neural Network is proposed based on the context incongruity hypothesis of Joshi et al. (2017), where sarcasm emerges from a clash between what is literally said and how it emotionally feels. DMS-CNN forces the network to model these two dimensions through separate, parallel input channels of semantic and sentiment.

Preprocessing. Text is tokenized using the NLTK TweetTokenizer, since the data originates from social media, standard tokenizers built for formal prose would destroy task-relevant signals. TweetTokenizer could handle capitalization patterns, such as “GREAT job” serve as informal markers of ironic tone, hashtags that are semantically meaningful, and elongated words like “sooooo” that carry intensity cues. All sequences are padded or truncated to a fixed maximum length of 60 tokens to enable uniform batch processing.

Channel A — Semantic Representation. Each token is mapped to a 100-dimensional embedding vector, initialized randomly and updated during training. This channel encodes the objective, denotative content of the utterance.

Channel B — Sentiment Representation. For each token, VADER computes a 3-dimensional polarity vector consisting of its positive, neutral, and negative scores. This produces a fixed sentiment feature sequence of shape (seq_len, 3) that encodes how each word feels in terms of lexical sentiment. This channel is designed to force the network to explicitly attend to polarity patterns at every token position.

Multi-Scale Convolution. Both channels are independently processed by three parallel 1D convolutional layers with kernel sizes of 2, 3, and 5, each producing 64 feature maps with ReLU activation and same-padding. The motivation follows Kim (2014) where different kernel sizes act as n-gram detectors at different scales, capturing bigram-level abrupt conflicts (e.g., “brilliant disaster”), phrase-level irony, and slightly longer structural setups respectively. The six resulting feature maps (3 scales $\times$ 2 channels) are concatenated along the filter dimension, yielding a combined representation of shape (seq_len, 384).

Self-Attention Mechanism. A single-head self-attention layer computes a scalar attention score for each position via a linear projection followed by softmax normalization. The output is a weighted sum of the 384-dimensional position features, producing a single context vector.

Classification. The context vector is passed through a dropout layer ($p = 0.3$) and a linear classifier with sigmoid activation, producing a probability score for the sarcastic class. Binary cross-entropy loss is used during training with the Adam optimizer ($lr = 0.001$, $weight_decay = 1e-4$). The model is trained for up to 10 epochs with early stopping (patience=3), saving the checkpoint with the highest validation F1 on the sarcastic class.

4.2 BERT - CNN

To investigate whether the performance of our multi-scale convolutional architecture can be improved through deeper semantic representations, we propose a BERT-CNN hybrid model.

The preprocessing pipeline shifts from word-level tokenization to subword tokenization using BertTokenizerFast from the Hugging Face Transformers library [8]. We use the bert-base-uncased vocabulary because subword tokenization is particularly effective for social media data, where misspellings and elongated words such as “soooo” are common. This approach reduces the likelihood that such words are treated as unknown tokens.

Overall, the architecture retains the dual-channel and multi-scale structural design of the DMS-CNN, with one critical upgrade to the semantic representation component:

1. Semantic Channel A (Frozen BERT):

This is the primary distinction from the DMS-CNN. Instead of training shallow word embeddings from scratch, Channel A uses a pre-trained BERT model. Because our dataset is relatively small, using a pre-trained model allows the channel to capture richer semantic information. By extracting the last hidden state from BERT, the model obtains a deep 768-dimensional contextualized

representation for each token. We freeze the BERT weights using PyTorch [7]. This design choice ensures that BERT functions purely as a static feature extractor, allowing us to isolate and evaluate the learning capacity of our custom CNN head.

2. Sentiment Channel B (VADER):

The remainder of the architecture follows the same strategy as the DMS-CNN, with Channel B preserving the original sentiment-based representation using VADER.

4.3 BERT

While the BERT-CNN model passively utilizes pre-trained representations as a frozen feature extractor alongside manually constructed sentiment features, our third model was developed to evaluate the effectiveness of pure end-to-end representation learning.

For this approach, we use the standard BertForSequenceClassification module based on BERT (bert-base-uncased) provided by Hugging Face Transformers[8]. In this architecture, we remove both the VADER sentiment features (Channel B) and the multi-scale CNN classification head. Unlike the BERT-CNN model, where the language model weights remain frozen, all parameters in this model are updated during training. This allows the internal attention heads to adapt dynamically to the sarcasm detection task and shift their focus according to task-specific semantic patterns. Training uses a learning rate of $2e-5$, linear warmup over 6% of total steps, and a weight decay of 0.01. The model is trained for 2 epochs with a batch size of 8, until the loss stops decreasing.

4.4 DeBERTa

Deberta extends BERT where its disentangled attention mechanism separately encodes the content and relative position of each token, computing attention weights using disentangled matrices for content-to-content, content-to-position, and position-to-content interactions, allowing the model to better leverage positional context when modeling the relationship between words, which is important for detecting sarcasm where the ironic signal often emerges from how semantically incongruous words are positioned relative to each other.

Setup. The model is initialized from the deberta-v3-base. All parameters are updated end-to-end during fine-tuning. Text is tokenized using DebertaV2TokenizerFast with a maximum sequence length of 128 tokens. Training uses the AdamW optimizer with a learning rate of $2e-5$, linear warmup over 6% of total steps, weight decay of 0.01, and gradient clipping at 1.0. The model is trained for 5 epochs with a batch size of 16, with no manual sentiment features or architectural modifications, therefore the model learns sarcasm representations entirely through end-to-end attention over the raw text.

5 Evaluation and Results

All models are evaluated on the same test set of 1,844 instances (926 non-sarcastic, 918 sarcastic), which comprises 1,333 instances from GEN, 207 from HYP, and 304 from RQ, reflecting the relative size of each subset in the full corpus. Accuracy, precision, recall, and macro-averaged F1-score is reported across all models.

5.1 Overall Results

DeBERTa achieves the highest performance across all metrics. Notably, our custom DMS-CNN (68.8%) underperforms the far simpler TF-IDF+SVM baseline (74.0%). Each step toward deeper contextual modeling, from frozen BERT features (BERT-CNN, 76.3%) to end-to-end fine-tuning (BERT, 79.4%) to disentangled attention (DeBERTa, 82.2%), produces consistent gains.

Table 1: Full Model Performance Comparison

Model	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)
Random Baseline	53.3%	0.53	0.53	0.53
TF-IDF + SVM	74.0%	0.74	0.74	0.737
DMS-CNN	68.8%	0.71	0.70	0.683
BERT-CNN (Hybrid)	76.3%	0.77	0.76	0.760
BERT (fine-tuned)	79.4%	0.80	0.79	0.794
DeBERTa (fine-tuned)	82.2%	0.82	0.82	0.822

Table 2: Per-Subset Macro F1 by Model

Model	GEN (n=1333)	HYP (n=207)	RQ (n=304)
TF-IDF + SVM	0.740	0.647	0.789
DMS-CNN	0.713	0.550	0.639
BERT-CNN (Hybrid)	0.755	0.728	0.805
BERT	0.799	0.729	0.815
DeBERTa	0.822	0.807	0.832

5.2 Per-Subset Results

UCSC Sarcasm Corpus is divided into three subsets: general sarcasm (GEN), hyperbole (HYP), and rhetorical questions (RQ). Table 3 shows how each model performs on these subsets separately.

HYP is consistently the hardest subset for every model, with all models achieving their lowest per-subset scores there, where DMS-CNN’s collapse on HYP (55.0%, barely above random) is the starkest failure point. Yet DeBERTa is the only model to surpass 80% on HYP, achieving 80.7% with a 7.8-point

improvement over BERT’s 72.9% on the same subset, despite only a 2.8-point gap between the two models on GEN.

6 Discussion

6.1 Finding 1: VADER harms performance, especially on hyperbole

The most counterintuitive result is that DMS-CNN (68.8%) underperforms TF-IDF+SVM (74.0%), despite being a substantially more complex architecture. The damage is most severe on HYP, where DMS-CNN drops to 55.0% while TF-IDF+SVM still got 64.7%.

This might be because hyperbolic sarcasm works by dramatically overstating a negative sentiment using positive or extreme language: “what?!?!?! blasphemy!!!! where would i get my cd fodder from?” and “I can’t believe that you would attempt to make such an insane assertion.” These sentences are sarcastic precisely because the emotional intensity of the language is disproportionate to the actual target. VADER could assign strong negative scores to words like “blasphemy” or “insane” in isolation, but in hyperbolic sarcasm, these words are often used performatively, not descriptively. The static lexicon cannot distinguish between genuine outrage and performed outrage, and since hyperbole often deploys extreme language as its primary flag, VADER’s token-level scores actively mislead Channel B.

This failure is in contrast with the existing literature. Poria et al. (2016) demonstrated that CNNs can successfully exploit pre-extracted sentiment and emotion features for sarcasm detection, but their features were derived from pre-trained deep networks rather than lexicon lookups, giving them implicit contextual grounding. Our experiment shows that substituting a context-free lexicon for a context-aware sentiment extractor in this type of dual-channel architecture is not a minor degradation but a catastrophic one, particularly for sarcasm types that weaponize extreme language as their primary vehicle. Similarly, BNS-Net (Zhou et al., 2023) achieves strong results with its dual-channel design, but its Sentence-level Conflict Channel operates on learned sentence representations rather than static lexicon scores, a distinction our results suggest is fundamental rather than incidental.

6.2 Finding 2: Pretrained contextual representations perform better, especially on the hardest subset.

Comparing BERT-CNN (76.3%) with DMS-CNN (68.8%) isolates the contribution of pretrained representations: both use the same multi-scale CNN classification head, but BERT-CNN replaces randomly initialized embeddings with frozen BERT contextual representations. The gain is substantial, especially on HYP, it is a dramatic 17.8 points.

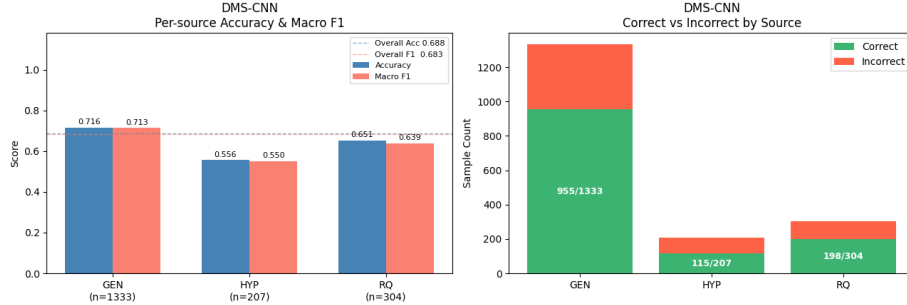


Figure 1: DMS-CNN Per-Source Accuracy

The HYP result is particularly revealing. Hyperbolic sarcasm like “that sounds a lot like another species... what is it called... oh, right, humans!” relies on an extended rhetorical buildup where the sarcastic punch emerges from the accumulated context across the sentence. A 100-dimensional embedding trained from scratch on 7,400 sentences cannot learn this kind of long-range contextual irony. BERT’s representations, having been pretrained, already encode the pragmatic implausibility of comparing humans to dangerous animals as a rhetorical device, which allows even the frozen features to support far better classification of hyperbolic instances.

This finding is related to the representational assumptions in Kim’s (2014) TextCNN, where Kim showed that even simple convolutional architectures perform substantially better with pre-trained word2vec embeddings than with randomly initialized ones, and attributed this to the rich semantic geometry encoded in the pretrained space. Our experiment pushes this finding further: the quality gap between a task-specific lexicon (VADER) and large-scale contextual pretraining (BERT) is far larger than the gap between random initialization and static word vectors, and this gap widens precisely on the sarcasm types that most require understanding pragmatic context beyond individual token meanings. Hyperbolic sarcasm relies on an extended rhetorical buildup where the ironic force accumulates across the sentence. BERT’s representations, since having been exposed to billions of tokens of human language, carry implicit knowledge of when such comparisons are being used ironically.

6.3 Finding 3: DeBERTa’s disentangled attention provides the largest gain on HYP.

The gap between BERT-CNN (76.3%, frozen BERT + VADER) and BERT fine-tuned (79.4%, no VADER) indicates that unfreezing BERT allows its attention heads to actively adapt to sarcasm detection, and removing VADER eliminates the harmful static sentiment signal.

Noted that while BERT achieves 72.9% on HYP, DeBERTa jumps to 80.7%, a 7.8-point improvement that is nearly three times larger than DeBERTa’s ad-

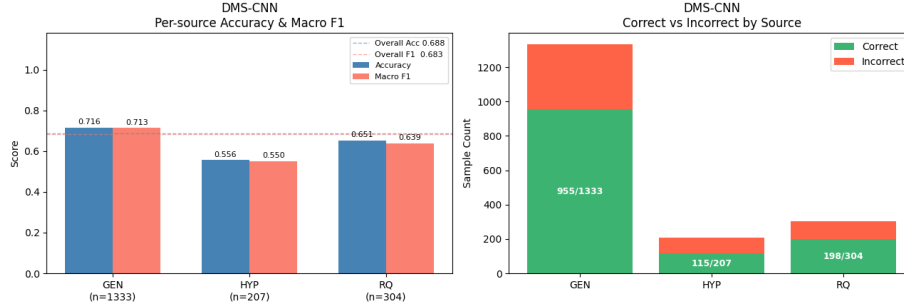


Figure 2: DeBERTa Per-Source Accuracy

vantage over BERT on GEN (2.3 points) or RQ (1.6 points), suggesting that DeBERTa’s disentangled attention mechanism, where it separately encodes content and relative positional relationships, provides a specific advantage for hyperbole detection. This might be because hyperbolic sarcasm frequently involves a structural contrast between an opening emotional claim and a deflating conclusion, or between the register of language used and the banality of the actual situation being described. And DeBERTa’s modeling of position-to-content attention weights may allow it to more precisely locate and weight these structural contrasts, whereas BERT’s standard self-attention conflates positional and content signals.

This result is in contrast with previous literature, rather than manually designing architectural channels to capture incongruity, as the dual-channel literature (Azwar & Suharjito, 2020; Zhou et al., 2023) has pursued, the disentangled attention in DeBERTa may be implicitly learning something structurally similar, separately tracking what is said where, and detecting the mismatch between those two dimensions, in an even more effective way.

RQ instances like “So there is nothing that can be called an absolute improvement? Well that certainly drops the chances of evolution producing what we see here today even further” exhibit a clear question-then-ironic-answer structure. All models perform relatively well on RQ (BERT: 81.6%, DeBERTa: 83.2%), suggesting that the structural regularity of the rhetorical question format is learnable from surface patterns even without deep pragmatic modeling.

7 Conclusion

This project investigates whether it is better to explicitly engineer architectural inductive biases for sarcasm detection, or to rely on the implicit contextual representations learned by large pretrained language models? Motivated by Joshi et al.’s (2017) context incongruity theory and the dual-channel architectures of Poria et al. (2016) and Zhou et al. (2023), DMS-CNN is designed to explicitly force the network to model the clash between semantic content and sentiment polarity.

The result is that implicit contextual modeling is more effective in sarcasm detection, where DeBERTa achieves 82.2% macro F1 compared to 68.8% for DMS-CNN, a gap that is larger than the gap between DMS-CNN and the random baseline. The most instructive result is the underperformance of DMS-CNN relative to TF-IDF+SVM (74.0%). We built an architecture specifically designed to detect sentiment incongruity, and it performed worse than a model with no sentiment modeling. The per-subset analysis indicates that on hyperbolic sarcasm (HYP), where extreme language is deployed performatively rather than descriptively, VADER assigns confident and systematically wrong sentiment scores, converting the intended signal of irony into noise.

The results show that when building sarcasm detection for informal text, pretrained language model fine-tuning is more effective than architectural complexity. The dual-channel incongruity framework in the previous literature depends entirely on whether the sentiment channel can perceive context.

8 Other Things We Tried

Our initial DMS-CNN implementation without regularization showed severe overfitting: by epoch 10, training accuracy reached 97.6% while validation accuracy dropped to 67.0%. We added early stopping with patience=3 and L2 regularization (weight_decay=1e-4), which caused training to terminate at epoch 6 and improved final test accuracy from 67% to 68.8%.

9 What We Would Have Done Differently or Next

Firstly, we would reconsider the use of VADER as the sentiment channel. We adopted it because it provides an explicit, interpretable, token-level sentiment signal directly aligned with Joshi et al.’s (2017) context incongruity framework. Our results demonstrate that this was less effective since VADER’s context-independence is not a minor limitation but a fundamental incompatibility with a task defined by context-dependent meaning reversal. This is ironic because model is used to detect the gap between literal and intended meaning, yet populates the literal-meaning channel with a tool that cannot perceive any gap at all.

Secondly, we would have pursued a more systematic error analysis using the per-subset breakdown as a starting point. The 7.8-point DeBERTa advantage over BERT specifically on HYP could have more investigation, such as which specific hyperbolic constructions does DeBERTa correctly classify that BERT misses? Are they characterized by particular structural features that DeBERTa’s disentangled positional attention can locate more precisely? Answering these questions would move the analysis from aggregate performance comparison toward a linguistically grounded account of what the models have actually learned about the pragmatics of irony, and would speak directly to the

open question in the literature of whether architectural inductive biases for incongruity detection (as pursued by Joshi et al., Azwar & Suharjito, and Zhou et al.) remain valuable in the era of large pretrained models, or whether those models have rendered explicit incongruity modeling out of date.

10 Group Effort

Feifan was in charge of data collection, preprocessing, implementing the baseline models, and implementing models. Yiyang was in charge of implementing the baseline models, designing the DMS-CNN model, implementing models and visualization.

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